Model

We use an IVQR Monte Carlo design with one endogenous treatment, one excluded instrument, and high-dimensional controls:

$$Y = \alpha_0 D + g_\tau(X) + U_\tau, \qquad D = \pi Z + h(X) + V.$$

Final design grid:

Component	Values
DGPs	DGP1 baseline sparse, DGP2 denser sparse, DGP3 heavy-tail sparse
Active controls	DGP1: 5, DGP2: 10, DGP3: 5
Sample sizes	$n \in \{500, 1000\}$
Controls	$p \in \{200, 500\}$
Instrument strength	$\pi \in \{1.00, 0.50, 0.25, 0.10\}$
Quantiles	$\tau \in \{0.25, 0.50, 0.75\}$
Base seed	12345

Estimators:

Estimator	Role
Oracle IVQR	Benchmark estimator using the true active controls.
Post-selection IVQR	Feasible sparse estimator using Lasso-based control selection.
DML-IVQR	Cross-fitted DML-style IVQR estimator with quantile-regression nuisance fitting.

Experiment 1: R = 10 runs

Metric	Oracle IVQR	Post-selection IVQR	DML-IVQR
Coverage (%)	86.04	76.74	**87.99**
CR length	2.148	**1.849**	2.642
RMSE	**0.846**	0.893	0.986
MAE	**0.607**	0.647	0.753
Mean runtime (s)	**0.49**	13.68	27.39

The first run showed two problems: post-selection undercovered strongly, and DML-IVQR was already computationally expensive. Oracle coverage was also below nominal, which suggested finite-sample and weak-IV difficulty rather than only an implementation problem.

We then corrected the final sparsity design (DGP1 and DGP3 use s=5, DGP2 uses s=10) and reran the same estimator comparison.

Metric	Oracle IVQR	Post-selection IVQR	DML-IVQR
Coverage (%)	**90.00**	81.46	88.26
CR length	2.269	**2.029**	2.584
RMSE	0.860	**0.847**	1.024
MAE	0.613	**0.610**	0.783
Mean runtime (s)	**1.93**	4.85	82.40

The sparsity correction improved Oracle and post-selection coverage, but post-selection still undercovered. DML-IVQR remained statistically stable but became too slow in the full sparse design.

Experiment 2: Critical-value multiplier

Metric	CV 1.0	CV 1.1	CV 1.2
Oracle coverage (%)	86.04	87.01	**87.57**
Oracle CR length	**2.148**	2.217	2.279
Post-selection coverage (%)	76.74	77.78	**78.61**
Post-selection CR length	**1.849**	1.909	1.964

Increasing the critical-value multiplier produced only small coverage gains. Therefore the critical-value multiplier was kept at 1.0, and the focus moved to estimator-specific tuning.

Experiment 3: Oracle IVQR R100 grid check

Metric	Grid21: [-1,3]	Grid41: [-2,4]
Coverage (%)	**90.00**	89.74
CR length	**2.263**	2.875
RMSE	**0.830**	1.019
MAE	**0.599**	0.695

So grid21 means: 21 alpha values with step 0.2 on the interval [-1, 3]. Grid41 means: 41 alpha values with step 0.15 on the wider interval [-2, 4]. The wide grid did not increase coverage, but it reduced boundary truncation.

Experiment 4: Post-selection IVQR R20. Lasso selection multiplier

Multiplier	Coverage (%)	CR length	RMSE	MAE	Mean runtime (s)
Mult. 1.0	81.46	**2.029**	0.847	0.610	**4.85**
Mult. 1.2	86.46	2.166	0.854	0.614	6.63
Mult. 1.5	88.51	2.253	0.846	0.608	5.39
Mult. 1.7	90.10	2.275	0.844	**0.604**	14.94
Mult. 1.8	90.31	2.279	**0.843**	**0.604**	5.37
Mult. 2.0	**90.97**	2.290	**0.843**	0.607	6.78

The true average sparsity across the three DGPs is about 6.67 controls. The baseline post-selection estimator selected too many controls, about 19.1 on average, and undercovered. Increasing the Lasso multiplier reduced the selected set to 11.45 for multiplier 1.2, 7.36 for 1.5, 6.36 for 1.7, and 6.07 for 1.8.

Multiplier 1.8 is kept as the main choice because it balances coverage, accuracy, selection size, and runtime. Multiplier 2.0 gives slightly higher coverage but is more aggressive and is kept as a robustness specification.

Experiment 5: Alpha-grid experiment with multiplier 1.8

Metric	Narrow [0,2], 21	Default [-1,3], 21	Wide [-2,4], 41
Coverage (%)	**90.76**	90.31	90.21
CR length	**1.397**	2.279	2.890
RMSE	**0.584**	0.843	1.052
MAE	**0.431**	0.604	0.713
Mean runtime (s)	9.40	**5.37**	6.66

The narrow grid gives mechanically shorter confidence regions and smaller RMSE/MAE because it truncates the search interval, so it is not suitable as final evidence. The wide grid reduces boundary truncation but increases interval length and cost. For final reporting, we use the default grid21 (-1,3) for all main estimators to keep results comparable across Oracle, post-selection, and DML-IVQR.

Experiment 6: DML-IVQR. Quantile-regression penalty

Metric	Pen. 0.01	Pen. 0.02	Pen. 0.05	Pen. 0.07	Pen. 0.10
Coverage (%)	85.00	92.50	94.17	**96.67**	**95.83**
CR length	**2.635**	2.775	2.994	3.060	3.133
RMSE	1.078	**0.956**	1.033	0.993	1.053
MAE	0.826	**0.745**	0.789	0.807	0.816
Mean runtime (s)	100.62	80.15	53.69	**34.09**	53.18

The DML-IVQR problem was dense quantile nuisance fitting: with penalty 0.01 the nuisance models were too large and slow. Penalty 0.07 gave the best balance in the calibration subset: strong coverage, lower runtime, and much sparser nuisance fits.

Metric	highs-ds	highs-ipm
Coverage (%)	**85.00**	**85.00**
Mean runtime (s)	178.20	**100.60**
Median runtime (s)	135.50	**79.20**
P95 runtime (s)	325.80	**162.50**
Max runtime (s)	345.10	**166.90**

`highs-ipm` gave the same coverage as `highs-ds` in the solver comparison but was substantially faster. Therefore, `highs-ipm` was chosen for the final DML-IVQR runs.

Experiment 7: Folds and grid size

Setting	Coverage (%)	CR length	RMSE	MAE	Mean runtime (s)
K2, grid21	93.33	3.090	1.084	0.840	39.62
K3, grid11	95.00	3.051	1.054	0.850	**30.62**
K3, grid15	98.33	3.078	1.042	0.847	45.93
K3, grid21	**96.67**	3.060	**0.993**	0.807	34.09
K5, grid21	95.83	**2.947**	0.994	**0.743**	109.33

K2 lowered cross-fitting but lost coverage, while K5 was much slower and did not justify its cost. Smaller grids gave only moderate runtime savings.

Experiment 8: DML-IVQR R20 full-design confirmation

Metric	K3, pen. 0.01	K3, pen. 0.07
Coverage (%)	88.16	**95.52**
CR length	**2.593**	2.960
RMSE	1.021	**0.983**
MAE	0.777	**0.757**
Mean runtime (s)	59.34	**31.63**

The full-design R20 run confirmed that the tuned DML-IVQR setting is not only a calibration improvement. It raised coverage close to nominal while cutting average runtime almost in half.

Experiment 9: Replication depth

Run	Coverage (%)	CR length	RMSE	MAE	Mean runtime (s)
Oracle R10	90.00	2.269	0.860	0.613	**1.93**
Oracle R100	90.00	**2.263**	**0.830**	**0.599**	–
Post-selection R20, mult. 1.8	90.31	2.279	0.843	0.604	5.37
DML-IVQR R20, final	**95.52**	2.960	0.983	0.757	31.63

The R100 Oracle run confirms that the initial R10 Oracle result was stable. Post-selection and DML-IVQR were tuned with R20 runs because they are more computationally expensive.

Final decisions

Final DML-IVQR choice: use K=3, quantile penalty 0.07, solver highs-ipm, ridge alpha 1.0, and grid [-1, 3] with 21 points.

Component	Final choice
Alpha grid	Default grid21: [-1, 3], 21 points, step 0.2
Critical-value multiplier	1.0
Oracle IVQR	True active controls
Post-selection IVQR	Lasso selection multiplier 1.8
DML-IVQR	K=3, quantile penalty 0.07, solver highs-ipm, ridge alpha 1.0

highs-ipm = HiGHS Interior Point Method